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DB Schema

19 tables · 6 sections

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about schemas :

This schema covers 6 areas of the hospital system.

- auth & users — `role`, `user`
- patients — `patient`, `patient_insurance`, `patient_insurance_documents`
- doctors — `doctor`, `doctor_availability`, `appointment`
- wards — `ward`, `bed`, `admission`, `discharge_summary`
- billing — `service`, `ledger`, `ledger_entry`, `insurance`, `payment`
- inventory — `inventory_item`, `purchase_request`
- `ledger_entry` cannot exist without a `ledger` (weak entity).

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section 1 — auth & users

1. role

Defines the type of user in the system.

```
role (
  role_id      int          primary key,
  role_name    varchar(50)  unique not null
)
```

relationships:

- `role` → `user` — 1 to n

participation:

- `user` — total
- `role` — partial

---

2. user

Every person who logs into the system.

```
user (
  user_id        int           primary key,
  name           varchar(100),
  email          varchar(100)  unique,
  password_hash  varchar(255),
  role_id        int,           -- fk → role
  created_at     timestamp
)
```

relationships:

- `user` → `patient` — 1 to 0..1
- `user` → `purchase_request` — 1 to n

participation:

- `patient` — total
- `user` — partial

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section 2 — patients

3. patient

Core patient profile.

```
patient (
  patient_id                int           primary key,
  user_id                   int           not null,  -- fk → user
  uhid                      varchar(20)   not null,
  name                      varchar(100)  not null,
  phone                     varchar(15),
  alternate_phone           varchar(15),
  dob                       date,
  gender                    varchar(10),
  blood_group               varchar(5),
  address                   text,
  emergency_contact_name    varchar(100),
  emergency_contact_phone   varchar(15),
  created_at                timestamp     default current_timestamp
)
```

relationships:

- `patient` → `appointment` — 1 to n

- `patient` → `admission` — 1 to 0..1
- `patient` → `patient_insurance` — 1 to 0..1
- `patient` → `discharge_summary` — 1 to n

participation:

- `Appointment` — total
- `patient` — partial

---

4. patient_insurance

Insurance policy linked to a patient.

```
patient_insurance (
  insurance_id   int           primary key,
  patient_id     int           unique not null,  -- fk → patient
  provider_name  varchar(100),
  policy_number  varchar(50),
  member_id      varchar(50),
  coverage_type  varchar(20),
  valid_from     date,
  valid_to       date,
  created_at     timestamp
)
```

relationships:

- `patient` → `patient_insurance` — 1 to 0..1
- `patient_insurance` → `patient_insurance_documents` — 1 to 0..3

participation:

- `patient_insurance` — total
- `patient` — partial

---

5. patient_insurance_documents

Uploaded insurance card images.

```
patient_insurance_documents (
  doc_id       int           primary key,
  insurance_id int,           -- fk → patient_insurance
  file_name    varchar(255),
  file_path    varchar(500),
  uploaded_at  timestamp
)
```

relationships:

- `patient_insurance` → `patient_insurance_documents` — 1 to 0..3

participation:

- `patient_insurance_documents` — total
- `patient_insurance` — partial

---

section 3 — doctors

6. doctor

Doctor details and specialization.

```
doctor (
  doctor_id       int           primary key,
  name            varchar(100),
  specialization  varchar(100),
  phone           varchar(15),
  email           varchar(100)
)
```

relationships:

- `doctor` → `doctor_availability` — 1 to n

---

7. doctor_availability

Doctor availability time slots.

```
doctor_availability (
  availability_id  int      primary key,
  doctor_id        int,      -- fk → doctor
  available_date   date,
  start_time       time,
  end_time         time,
  status           varchar(20)
)
```

relationships:

- `doctor` → `doctor_availability` — 1 to n
- `doctor_availability` → `appointment` — 1 to 0..n

---

8. appointment

Patient appointment booking.

```
appointment (
  appointment_id      int           primary key,
  patient_id          int           not null,  -- fk → patient
  availability_id     int           not null,  -- fk → doctor_availability
  scheduled_datetime  timestamp     not null,
  visit_type          varchar(20)   check (visit_type in ('opd', 'emergency', 'followup')),
  status              varchar(20)   check (status in ('pending', 'confirmed', 'cancelled',
'completed')),
  created_by          int,           -- fk → user
  created_at          timestamp     default current_timestamp
)
```

relationships:

- `patient` → `appointment` — 1 to n
- `doctor_availability` → `appointment` — 1 to n
- `appointment` → `admission` — 1 to 0..1

participation:

- `patient` — total
- `appointment` — total

---

section 4 — wards

9. ward

Hospital ward containing beds.

```
ward (
  ward_id     int          primary key,
  ward_name   varchar(50)  unique not null,
  total_beds  int          not null,
  description text
)
```

relationships:

- `ward` → `bed` — 1 to n

---

10. bed

Individual bed in a ward.

```
bed (
  bed_id      int          primary key,
```

```
  ward_id     int,          -- fk → ward
  bed_number  varchar(10),
  status      varchar(20)
)
```

relationships:

- `ward` → `bed` — 1 to n
- `bed` → `admission` — 1 to n

---

11. admission

Patient admission record.

```
admission (
  admission_id   int        primary key,
  appointment_id int,        -- fk → appointment
  patient_id     int,        -- fk → patient
  bed_id         int,        -- fk → bed
  admit_time     timestamp,
  discharge_time timestamp,
  status         varchar(20)
)
```

relationships:

- `appointment` → `admission` — 1 to 0..1
- `patient` → `admission` — 1 to 0..1
- `bed` → `admission` — 1 to n
- `admission` → `ledger` — 1 to 1
- `admission` → `insurance` — 1 to 0..1
- `admission` → `discharge_summary` — 1 to 1

---

12. discharge_summary

Patient discharge summary.

```
discharge_summary (
  summary_id      int            primary key,
  admission_id    int            unique,  -- fk → admission
  patient_id      int,            -- fk → patient
  discharge_notes text,
  final_amount    decimal(10,2),
  generated_at    timestamp,
  file_path       varchar(255)
)
```

relationships:

- `admission` → `discharge_summary` — 1 to 1
- `patient` → `discharge_summary` — 1 to n

articipation
ADMISSION (1) —— (1) DISCHARGE_SUMMARY
PATIENT (1) —— (N) DISCHARGE_SUMMARY

participation:

- `Admission(1)` — (1) total
- `patient(1)` — (N) partial

---

section 5 — billing

13. service

Hospital services and their costs.

```
service (
  service_id    int            primary key,
  service_name  varchar(100),
  base_cost     decimal(10,2)
)
```

relationships:

- `service` → `ledger_entry` — 1 to n
- `service` → `inventory_item` — 1 to n

---

14. ledger

Billing ledger per admission.

```
ledger (
  ledger_id    int        primary key,
  admission_id int        unique,  -- fk → admission
  status       varchar(20),
  created_at   timestamp
)
```

relationships:

- `admission` → `ledger` — 1 to 1

- `ledger` → `ledger_entry` — 1 to n
- `ledger` → `payment` — 1 to n

---

15. **ledger_entry** (weak entity)

Billable line items in a ledger.

```
ledger_entry (
  entry_id    int,
  ledger_id   int,            -- fk → ledger  (part of pk)
  service_id  int,            -- fk → service
  quantity    int,
  unit_price  decimal(10,2),
  amount      decimal(10,2),
  entry_time  timestamp,
  primary key (entry_id, ledger_id)
)
```

relationships:

- `ledger` → `ledger_entry` — 1 to n
- `service` → `ledger_entry` — 1 to n

participation:

- `ledger_entry` — total
- `ledger` — partial

---

16. **insurance**

Insurance claim record for an admission.

```
insurance (
  insurance_id         int            primary key,
  admission_id         int            unique not null,  -- fk → admission
  patient_insurance_id int            not null,         -- fk → patient_insurance
  coverage_percentage  decimal(5,2),
  claim_status         varchar(20)    check (claim_status in ('pending', 'approved',
'rejected'))
)
```

relationships:

- `admission` → `insurance` — 1 to 0..1

---

17. **payment**

Payment records for a ledger.

```
payment (
  payment_id    int            primary key,
  ledger_id     int,            -- fk → ledger
  amount_paid   decimal(10,2),
  payment_mode  varchar(20),
  payment_time  timestamp
)
```

relationships:

- `ledger` → `payment` — 1 to n

---

section 6 — inventory

18. inventory_item

Inventory stock tracking.

```
inventory_item (
  item_id         int          primary key,
  item_name       varchar(100),
  category        varchar(50),
  stock_quantity  int,
  reorder_level   int,
  service_id      int           -- fk → service
)
```

relationships:

- `inventory_item` → `purchase_request` — 1 to n
- `service` → `inventory_item` — 1 to n

---

19. purchase_request

Inventory restock request.

```
purchase_request (
  request_id          int        primary key,
  item_id             int,        -- fk → inventory_item
  quantity_requested  int,
  status              varchar(20),
  requested_by        int,        -- fk → user
  requested_at        timestamp
)
```

relationships:

- `inventory_item` → `purchase_request` — 1 to n
- `user` → `purchase_request` — 1 to n